# Metabolite Production

$$z = \frac{d[M]}{dt}$$

$$z = (v_{in} - v_{out}) \frac{1}{Vol}$$

$$\frac{\partial z}{\partial Vol} = -\frac{(v_{in} - v_{out})}{Vol^2}$$

$$\frac{\partial z}{\partial v_{in}} = \frac{1}{Vol}$$

$$\frac{\partial z}{\partial v_{in}} = -\frac{1}{Vol}$$

# Gradients

- $\left(\frac{-\frac{C_3 w}{x^2} - C_4}{\epsilon}\right)\hat{i} + \frac{C_3}{\epsilon x}\hat{j} - \left(\frac{w_2}{\exp(dw)^2}\frac{-1}{\exp(dw)} + \frac{2r^2 w}{3w^2 + w^2 r^{-2}}\right)\hat{k}$

- $C_1\hat{i} + \left(f'(y)\,T(t)y + T(t)\,f(y)\right)\hat{j}$

$$\frac{(v_R)_A - v_M}{v_M}$$

Unit vector

$$v_M \hat{i} + \left(\frac{(v_R)_A - v_M}{v_M}\right)\hat{j}$$

$$\sqrt{(v_M)^2 + \left(\frac{(v_R)_A - v_M}{v_M}\right)^2} \rightarrow \sqrt{(v_M)^2 + \left[\frac{(v_R)_A^2 - 2(v_R)_A v_M + v_M^2}{v_M^2}\right]}$$

$$\rightarrow \sqrt{(v_M)^2 + \left[\frac{(v_R)_A^2}{v_M^2} - \frac{2(v_R)_A \cancel{v_M}}{\cancel{v_M^2}\, v_M} + \frac{\cancel{v_M^2}}{\cancel{v_M^2}}\right]}$$

$$\rightarrow \sqrt{(v_M)^2 + \frac{(v_R)_A^2}{v_M^2} - \frac{2(v_R)_A}{v_M}}$$

$$\frac{v_M \hat{i} + \left(\frac{(v_R)_A - v_M}{v_M}\right)\hat{j}}{\sqrt{(v_M)^2 + \frac{(v_R)_A^2}{v_M^2} - \frac{2(v_R)_A}{v_M}}}$$

$$\left(\frac{1}{\sqrt{\mu_M^2 + \frac{(\mu_R)_A^2}{\mu_M^2} - \frac{2(\mu_R)_A}{\mu_M}}}\right)\left((\mu_M)\left(\frac{-c_3 w}{x^2} - c_4\right) + \left(\frac{(\mu_R)_A - \mu_M}{\mu_M}\right)\left(\frac{c_2}{\varepsilon x}\right)\right)$$

$$\left(-\frac{(V_{in} - V_{out})}{vol^2}\right)$$

Slope topology #1

$$\left(\frac{1}{\sqrt{\mu_M^2 + \frac{(\mu_R)_A^2}{\mu_M^2} - \frac{2(\mu_R)_A}{\mu_M}}}\right)\left((\mu_M)(c_1) + \left(\frac{(\mu_R)_A - \mu_M}{\mu_M}\right)\left(f'(u)T(t)_{dy} + T(t)f(u)\right)\right)$$

$$\left(-\frac{(v_{in} - v_{out})}{vol^2}\right)$$

Slope topology #2

# Topological Matrix Graph

| | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 1 | 1 | -1 | 0 | 0 | 0 | 0 |
| 2 | 1 | 0 | 0 | 0 | 0 | -1 |
| 3 | 0 | 1 | 0 | 1 | 0 | 0 |
| 4 | 0 | 0 | 0 | 1 | 1 | 0 |
| 5 | 0 | 0 | 0 | 0 | 1 | 1 |
| 6 | 0 | 1 | 0 | 0 | 0 | -1 |
| 7 | 0 | 1 | 1 | 0 | 0 | 1 |
| 8 | 0 | 0 | 1 | 0 | 0 | -1 |
| 9 | 0 | 0 | 1 | -1 | 0 | 0 |
| 10 | 0 | 1 | 0 | 0 | 1 | 0 |
| 11 | 0 | 0 | 1 | 0 | 1 | 0 |
| 12 | 0 | 1 | 1 | 1 | 0 | 0 |
| 13 | 1 | 0 | 1 | 0 | 0 | 0 |

| | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| | -1 | -1 | 0 | 0 | 0 | 0 |
| | 1 | 0 | 0 | 0 | 0 | -1 |
| | 0 | 1 | 0 | -1 | 0 | 0 |
| | 0 | 0 | 0 | -1 | -1 | 0 |
| | 0 | 0 | 0 | 0 | -1 | -1 |
| | 0 | 1 | 0 | 0 | 0 | -1 |
| | 0 | -1 | -1 | 0 | 0 | -1 |
| | 0 | 0 | 1 | 0 | 0 | -1 |
| | 0 | 0 | 1 | -1 | 0 | 0 |
| | 0 | -1 | 0 | 0 | -1 | 0 |
| | 0 | 0 | -1 | 0 | -1 | 0 |
| | 0 | -1 | -1 | -1 | 0 | 0 |
| | -1 | 0 | -1 | 0 | 0 | 0 |

$$
\begin{bmatrix}
x_1 - x_2 \\
x_1 - x_6 \\
x_2 + x_4 \\
x_4 + x_5 \\
x_5 + x_6 \\
x_2 - x_6 \\
x_2 + x_3 + x_6 \\
x_3 - x_6 \\
x_3 - x_4 \\
x_2 + x_5 \\
x_3 + x_5 \\
x_2 + x_3 + x_4 \\
x_1 + x_3
\end{bmatrix}
$$

$$
\begin{bmatrix}
-x_1 - x_2 \\
x_1 - x_6 \\
x_2 - x_4 \\
-x_4 - x_5 \\
-x_5 - x_6 \\
x_2 - x_6 \\
-x_2 - x_3 - x_6 \\
x_3 - x_6 \\
x_3 - x_4 \\
-x_2 - x_5 \\
-x_3 - x_5 \\
-x_2 - x_3 - x_4 \\
-x_1 - x_3
\end{bmatrix}
$$

# GNN Topology

$$\left(\frac{1}{\sqrt{\frac{\partial H}{\partial t}^2 + \frac{(\partial x)_A^2}{\partial H^2} + \frac{2(\partial x)_A}{\partial H}}}\right)\left(\left(\frac{-c_{gw}}{\varepsilon x^2} - \frac{c_{ik}}{\varepsilon}\right)\left(\frac{\partial H}{\partial t}\right) + \left(\frac{(\partial x)_A - \partial H}{\partial H}\right)\left(\frac{c_s}{\varepsilon x}\right)\right)\left(\frac{-(v_{in} - v_{out})}{vol^2}\right)$$

$$\begin{bmatrix} x_1 - x_2 \\ x_1 - x_6 \\ x_2 + x_4 \\ x_4 + x_5 \\ x_5 + x_6 \\ x_2 - x_6 \\ x_2 + x_3 + x_6 \\ x_3 - x_6 \\ x_3 - x_4 \\ x_2 + x_5 \\ x_3 + x_5 \\ x_2 + x_3 + x_4 \\ x_1 + x_3 \end{bmatrix}$$

$$\left(\frac{1}{\sqrt{\frac{\partial H}{\partial t}^2 + \frac{(\partial x)_A^2}{\partial H^2} + \frac{2(\partial x)_A}{\partial H}}}\right)\left(\left(\frac{-c_{gw}}{\varepsilon x^2} - \frac{c_{ik}}{\varepsilon}\right)\left(\frac{\partial H}{\partial t}\right) + \left(\frac{(\partial x)_A - \partial H}{\partial H}\right)\left(\frac{c_s}{\varepsilon x}\right)\right)\left(-\frac{(v_{in} - v_{out})}{vol^2}\right)$$

$$\begin{bmatrix} -x_1 - x_2 \\ x_1 - x_6 \\ x_2 - x_4 \\ -x_4 - x_5 \\ -x_5 - x_6 \\ x_2 - x_6 \\ -x_2 - x_3 - x_6 \\ x_3 - x_6 \\ x_3 - x_4 \\ -x_2 - x_5 \\ -x_3 - x_5 \\ -x_2 - x_3 - x_4 \\ -x_1 - x_3 \end{bmatrix}$$

$$\left(\frac{1}{\sqrt{dM^2+\frac{(dR)_A^2}{dM^2}+\frac{2(dR)_A}{dM}}}\right)\left((dM)(C_1)+\left(\frac{(dR)_A-dM}{dM}\right)\left(f'(y)T(x)_{dy}+T(x)f(y)\right)\right)\begin{bmatrix} x_1-x_2 \\ x_1-x_6 \\ x_2+x_4 \\ x_4+x_5 \\ x_5+x_6 \\ x_2-x_6 \\ x_2+x_3+x_6 \\ x_3-x_6 \\ x_3-x_4 \\ x_2+x_5 \\ x_3+x_5 \\ x_2+x_3+x_4 \\ x_1+x_3 \end{bmatrix}$$

$$\left(\frac{1}{\sqrt{dM^2+\frac{(dR)_A}{dM^2}+\frac{2(dR)_A}{dM^2}}}\right)\left((dM)(C_1)+\left(\frac{(dR)_A-dM}{dM}\right)\left(f'(y)T(x)_{dy}+T(x)f(y)\right)\right)\begin{bmatrix} -x_1-x_2 \\ x_1-x_6 \\ x_2-x_4 \\ -x_4-x_5 \\ -x_5-x_6 \\ x_2-x_6 \\ -x_2-x_3-x_6 \\ x_3-x_6 \\ x_3-x_4 \\ -x_2-x_5 \\ -x_3-x_5 \\ -x_2-x_3-x_4 \\ -x_1-x_3 \end{bmatrix}$$